# PULLBACK ALONG AN OPEN INCLUSION IS RESTRICTION OF SECTIONS, WITH STALKS AND COUNIT IDENTITIES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Definition of the presheaf pullback j_pG(V) as a colimit over opens W of X containing V. invoked at proof lines 4694-4697. Fixes the colimit whose largest element is computed.

*Source reference.* Original section 008C: sources/stacks-sheaves/pinned-original-source.tex, source lines 2202–2202. Cited at statement lines 4674-4675 for the definition of j_pG whose colimit is evaluated in the proof; load-bearing inside the counted span.

**Lemma 0.2.** *Let $X$ be a topological space. Let $j : U \to X$ be the inclusion of an open subset $U$ into $X$.*

1. *Let $\mathcal{G}$ be a presheaf of sets on $X$. The presheaf $j_p\mathcal{G}$ (see Section 008C) is given by the rule $V \mapsto \mathcal{G}(V)$ for $V \subset U$ open.*
2. *Let $\mathcal{G}$ be a sheaf of sets on $X$. The sheaf $j^{-1}\mathcal{G}$ is given by the rule $V \mapsto \mathcal{G}(V)$ for $V \subset U$ open.*
3. *For any point $u \in U$ and any sheaf $\mathcal{G}$ on $X$ we have a canonical identification of stalks*
$$j^{-1}\mathcal{G}_u = (\mathcal{G}|_U)_u = \mathcal{G}_u.$$
4. *On the category of presheaves of $U$ we have $j_p j_* = id$.*
5. *On the category of sheaves of $U$ we have $j^{-1}j_* = id$.*

*The same description holds for (pre)sheaves of abelian groups, (pre)sheaves of algebraic structures, and (pre)sheaves of modules.*

*Proof.* The colimit in the definition of $j_p\mathcal{G}(V)$ is over collection of all $W \subset X$ open such that $V \subset W$ ordered by reverse inclusion. Hence this has a largest element, namely $V$. This proves (1). And (2) follows because the assignment $V \mapsto \mathcal{G}(V)$ for $V \subset U$ open is clearly a sheaf if $\mathcal{G}$ is a sheaf. Assertion (3) follows from (2) since the collection of open neighbourhoods of $u$ which are contained in $U$ is cofinal in the collection of all open neighbourhoods of $u$ in $X$. Parts (4) and (5) follow by computing $j^{-1}j_*\mathcal{F}(V) = j_*\mathcal{F}(V) = \mathcal{F}(V)$.

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The exact same arguments work for (pre)sheaves of abelian groups and (pre)sheaves of algebraic structures. □